

SHARUNANDHAGANESH S

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CAREER OBJECTIVE

Driven Electronics and Communication Engineering graduate with a strong foundation in embedded systems, IoT architecture, and AI/ML — demonstrated through a research-grade capstone project (IS-TEWS) achieving AUC 0.9957 with a Zenodo-indexed published dataset. Cleared TCS NQT Digital category (Ref: DT20257137018), reflecting proven aptitude in programming and analytical problem-solving. Eager to apply hands-on hardware-software integration skills, a research mindset, and a commitment to continuous learning to contribute meaningfully in any engineering or technology-driven environment.

EDUCATION

B.E. – Electronics and Communication Engineering Francis Xavier Engineering College, Tirunelveli, Tamil Nadu CGPA: 7.9 (up to 7th Sem) No Active Backlogs Full-Time Course	2022 – 2026
Higher Secondary Certificate (HSC) – Vimal Matric Hr. Sec. School Kalugumalai, Tamil Nadu Percentage: 66.84%	2021 – 2022
Secondary School Leaving Certificate (SSLC) – Vimal Matric Hr. Sec. School Kalugumalai, Tamil Nadu Percentage: 80.1%	2019 – 2020

TECHNICAL SKILLS

Core Languages: C, Embedded C, Python
AI / ML & Data: LightGBM (GBDT), TensorFlow, Scikit-learn, Feature Engineering, Physics-Informed Inference Filtering, REST API (Flask)
Embedded Systems & IoT: ESP8266, Arduino UNO, Microcontroller Programming, Sensors & Actuators Interfacing, IoT Architecture, Real-time Data Acquisition, Embedded C Firmware
Web & Cloud Tools: Flask, Leaflet.js, Firebase, ThingSpeak, HTML, CSS, Git & GitHub
Testing & Hardware: Basic PCB Knowledge, Circuit Assembly, Soldering, SMT, Fault Debugging, Multimeter, Oscilloscope, Multisim, Tinkercad

INTERNSHIP EXPERIENCE

PCB Systems Intern – Vasantha Advanced System, Coimbatore
Industrial Internship | June 2024 – July 2024

- Assisted in circuit assembly, component selection, and quality inspection; contributed to assembling and testing over 20 production-ready electronic boards during the internship period.
- Conducted testing, debugging, and troubleshooting of electronic devices, helping reduce defect identification time through systematic fault analysis.
- Acquired hands-on exposure to SMT manufacturing operations and industrial production workflows, gaining practical understanding of quality compliance standards.

ACADEMIC PROJECTS

IS-TEWS — India-Specific Tsunami Early Warning System
LightGBM • Python • Arduino UNO • ESP8266 • Embedded C • Flask • Leaflet.js | 2025–2026 | DOI: 10.5281/zenodo.18901691

- Developed LightGBM v29 classifier (581 trees) trained on 17,038 USGS earthquake events with a 30-dimensional geophysical feature vector — achieved AUC 0.9957, Recall 0.8571, F1 0.8000; only 11 missed tsunamis and 22 false alarms on a 2,555-event test set covering 2019–2024.
- Implemented a four-rule physics-informed inference filter and built a Flask REST API (21 endpoints) with Leaflet.js interactive dashboard for real-time hazard visualization.
- Developed production ESP8266 firmware in Embedded C — Wi-Fi initialization, HTTP REST API calls, GPIO-driven multi-channel alerts (buzzer, LED, LCD), watchdog timer; alert latency under 2 seconds.
- Published dataset on Zenodo (DOI: 10.5281/zenodo.18901691), making the 17,038-event feature-engineered earthquake dataset publicly available for research reproducibility.

Industrial Energy Monitoring System
ESP8266 • Embedded C • ThingSpeak • Firebase | 2024

- Built an IoT-based real-time industrial energy monitoring system using ESP8266 with current and voltage sensors; programmed firmware in Embedded C for continuous data acquisition and transmission.
- Integrated ThingSpeak for real-time analytics dashboards and Firebase for persistent data storage, enabling remote monitoring of multiple industrial energy parameters simultaneously.

CERTIFICATIONS & ACHIEVEMENTS

- TCS NQT – Cleared Digital Category | Reference ID: DT20257137018 (March 2026)
- Dataset Published on Zenodo | DOI: 10.5281/zenodo.18901691 (2026)
- NPTEL – Industry 4.0 & Industrial Internet of Things | Score: 75% (Jul – Oct 2025)
- 4-Day Training: AI-ML in IoT using Python | Hackathon 1.0 (Aug 2023)

SOFT SKILLS

- Strong Verbal & Written Communication
- Analytical Thinking & Problem Solving
- Team Collaboration & Individual Contribution
- Adaptability & Continuous Learning